

LCM (Least Common Multiple)

Q Use the ladder method to find the LCM of the numbers 4, 8 and 10.

A $2 \mid 4 \quad 8 \quad 10$

Find a factor of *at least* two of the numbers. The more of the numbers your factor divides into, the more efficient the process. Write this factor in front of the three numbers, separated by a vertical line.

$2 \mid 4 \quad 8 \quad 10$
 $2 \quad 2 \quad 5$

Divide each number by the factor (**2**, in this example) and write the quotients below the original numbers.

$2 \mid 4 \quad 8 \quad 10$
 $2 \mid 2 \quad 4 \quad 5$
 $1 \quad 2 \quad 5$

Now find a factor (**2** again, here) of at least two of the new numbers and write it in front. If a number does not have that factor simply carry it down to the next row. (Here, 5 does not have 2 as a factor.)

$2 \mid 4 \quad 8 \quad 10$
 $2 \mid 2 \quad 4 \quad 5$
 $1 \quad 2 \quad 5$

Continue until no more factors can be found. Multiply all the **factors** and **remainders** together to find the lowest common multiple.

$$2 \times 2 \times 1 \times 2 \times 5 = 40$$

The LCM of 4, 8 and 10 is 40.



LCM (Ladder Method)

1 Find the LCM of each set of numbers.

a 4, 6

b 8, 12

c 12, 64

d 24, 32

e 35, 85

f 18, 54, 66

g 16, 20, 80

h 48, 68, 94

i 108, 180, 396

